

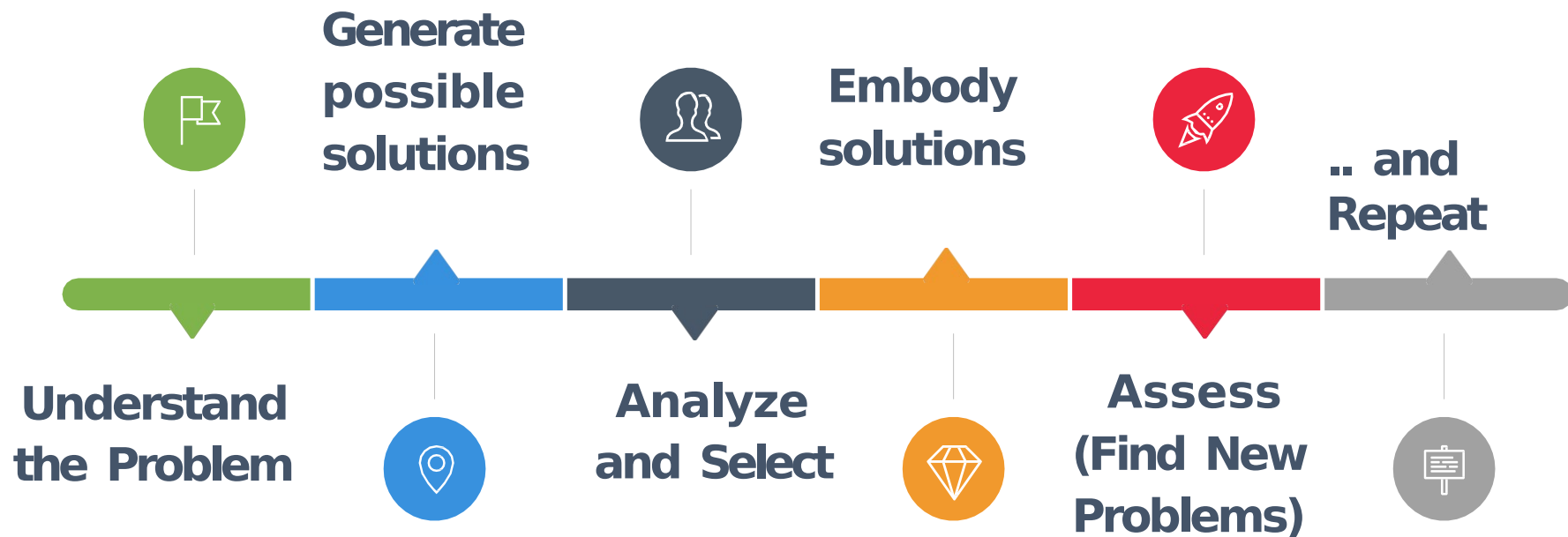
CSC-335

HCI & Computer

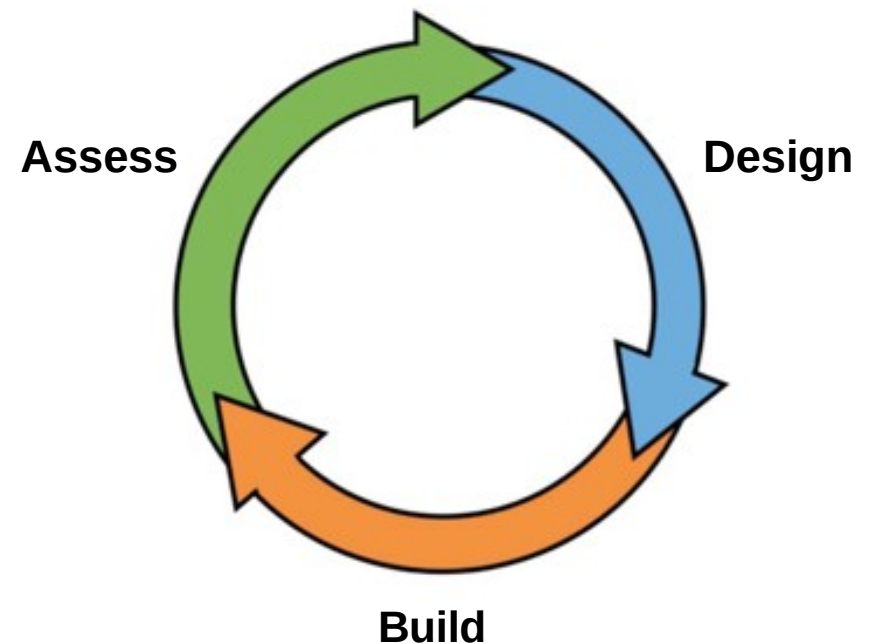
Graphics

Lec 7: Design, Prototyping and Sketching

The Design Process



Aligning The Design Process



What is special about UX Design

- Interactive systems are **time-based** -> experiences unfold over time
- Built on **action-response rules**: what users do, and how the system responds
- Designing actions = defining **what commands/options** users have
- Designing responses = deciding **what information** is presented back
- Key usability focus: helping users **navigate complex systems** to accomplish their goals

What is special about UX Design

Context is critical

- Who, how, when, and why: student at deadline vs admin during office hours at an LMS portal
- Other systems and devices: booking a seat for Eid?
- Other activities happening at the same time: divided attention e.g food panda, doctors
- Other people involved: banking app
- Actual user vs user on paper

The UX Design Process

- **Understand the problem**
 - **Generate possible solutions**
 - **Analyze and select Embodiment solutions**
 - **Assess (find new problems)**
 - **... and Repeat**
- **Study users: tasks, contexts, personas Sketch, storyboard, wireframes Apply UX criteria**
 - **Build prototype**
 - **Apply UX research Methods**
 - **... and Repeat**

Prototypes



Why do we prototype?

Why we prototype:

- turn abstract ideas into something concrete you can actually test
- step back and check if the design is meeting your goals or going off track
- give stakeholders something tangible to react to, so everyone's on the same page
- used in user testing to see if it actually works for real users

Prototypes can have many forms...

- A “representation of design” can mean a
 - *Verbal description*
 - *Sketch*
 - *Storyboard*
 - *Navigation/Flow diagram*
 - *Physical Model*
 - *Video*
 - *Formal specification*



In UX...

- In UX a prototype is a representation of the system as it will be experienced by end users.

A hand-drawn sketch of a mobile app screen titled "WELCOME!". Below the title is the text "Let's set up your profile.". The screen contains a profile icon with an "Edit" button, a "Name" label with a text input field, a "Gender" label with three radio button options labeled "OF", "OM", and "OO", a "Birthday (optional)" label with a date picker showing "JAN" and "1900", a "Phone Number (optional)" label with a text input field, and a "Let's Go" button at the bottom.

**Low
Fidelity**

A digital mockup of a mobile app screen titled "Crime Alert". Below the title is a "Report Crime" button. The screen contains a "Description" label with a text area containing the text "I witnessed a car getting broken into today. I have attached a picture of the car.", an "Address of Crime" label with a text input field containing "12", and a "Post" button at the bottom. The screen also displays two address suggestions: "123 Blue Road, Detroit, MI" and "1234 Victory Lane, Detroit, MI".

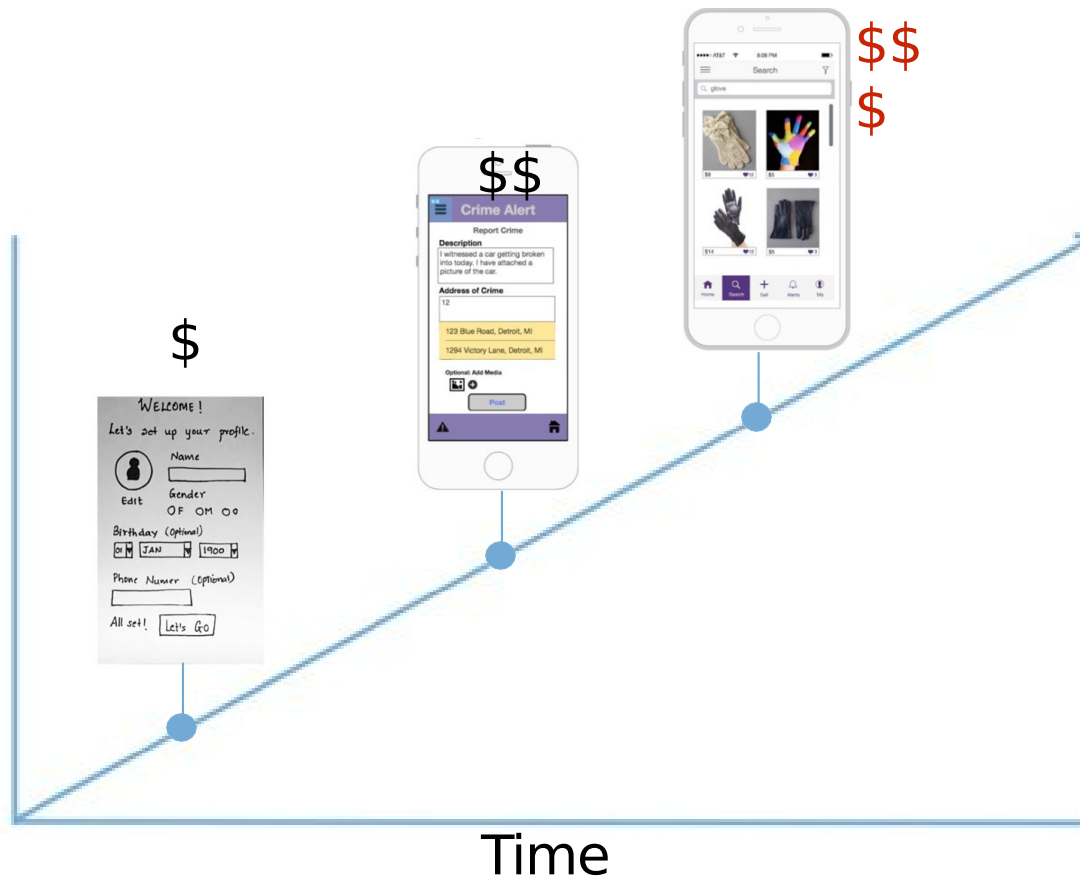
**Middle
Fidelity**



**High
Fidelity**

Progression of Prototypes

Fidelity Cost



Lo-Fi Prototypes

A hand-drawn Lo-Fi prototype of a user profile setup screen. The screen is titled "WELCOME!" and "Let's set up your profile." It features a circular profile icon with a person silhouette, labeled "Edit". Below the icon is a "Name" label and a text input field. To the right of the name field is a "Gender" label with radio buttons for "OF", "OM", and "OO". Below the gender section is a "Birthday (Optional)" label, followed by three dropdown menus showing "01", "JAN", and "1900". Below the birthday section is a "Phone Number (Optional)" label and a text input field. At the bottom, it says "All set!" followed by a "Let's Go" button. A blue circle is connected to the bottom of the screen by a vertical line.

- **Address**

- *Functionality*
- *Basic organization*
- *Task flow and coverage*

- **Ignore**

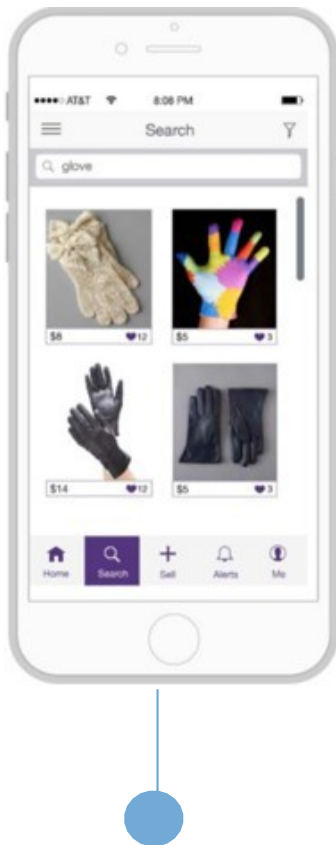
- *Graphics, fonts, colors*
- *Programming*
- *Real data*

Mid-Fi Prototypes



- **Address**
- *Layout*
 - how specifically are all of the different fields and controls going to be laid out?
- *Interactivity*
 - how does it actually feel to interact with the system?
- *Navigation*
 - And how do you navigate from one part of the system to another?
- **Ignore**
 - *Graphics, fonts, colors*
 - *Programming*
 - *Real data*

Hi-Fi Prototypes



- **Address:**

- *Graphic design*
- *Interaction details*
- *Realistic data*

- **Ignore**

- *Backend programming*
- *Complete functional coverage*
 - focused on the task that that particular prototype is supposed to accomplish

Avoid Premature Committment

- Investing resources (time & money) in design ideas that turn out to be a dead end.
- **Problems:**
 - *waste money and effort*
 - *exhausting the project resources, in running out of time,*
 - where we might end up being **stuck with a bad design** or design that isn't as optimal
 - very ***difficult to let go*** of those ***cognitive and emotional commitments*** that stakeholders make once they've invested significant time and resources into a particular design direction.

**“Low fidelity
prototypes are not
precious and
therefore invite
more honest
critique”**

-Bill Moggridge

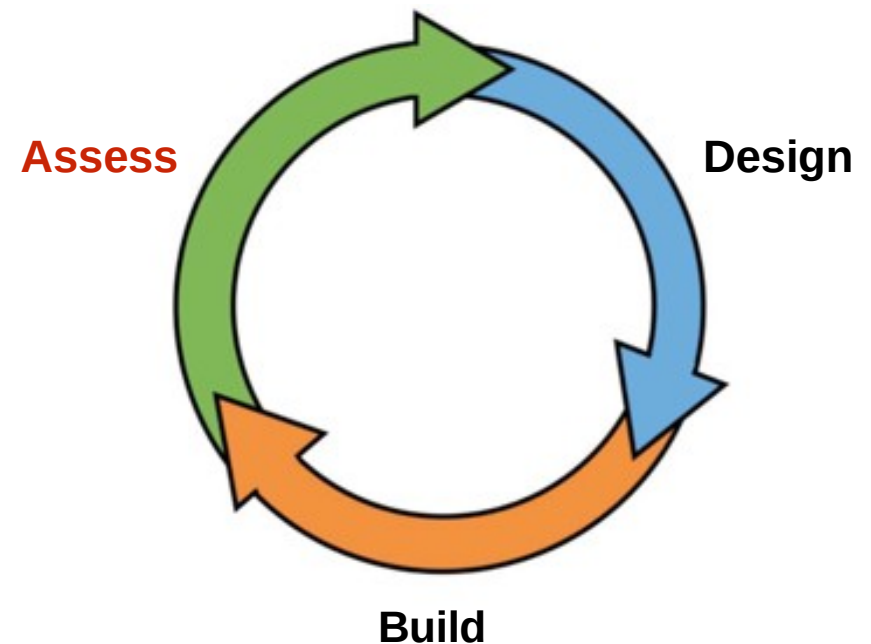


How does prototyping fit into the UX design process?



What happens before prototyping?

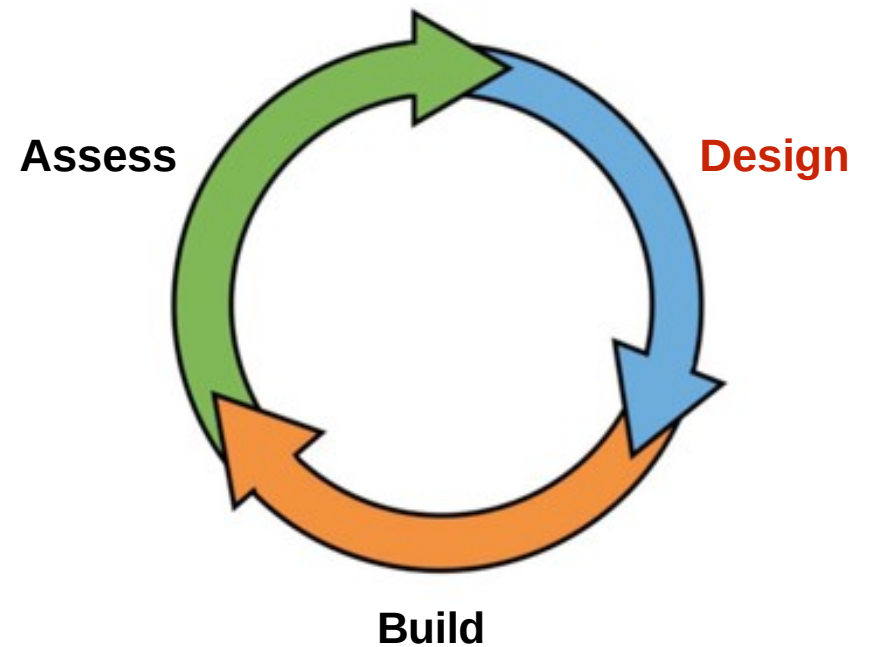
- **Assessing current needs**
 - *Understanding the design problem*
- **Establishing guidelines and constraints**
 - *what a good solution would look like*
- **Personas, scenarios, user stories, requirements**
 - *to guide design process*



What happens before prototyping?

- **Design!**

- *Ideation*
- *Sketching*
- *Selection*
- *Synthesis*



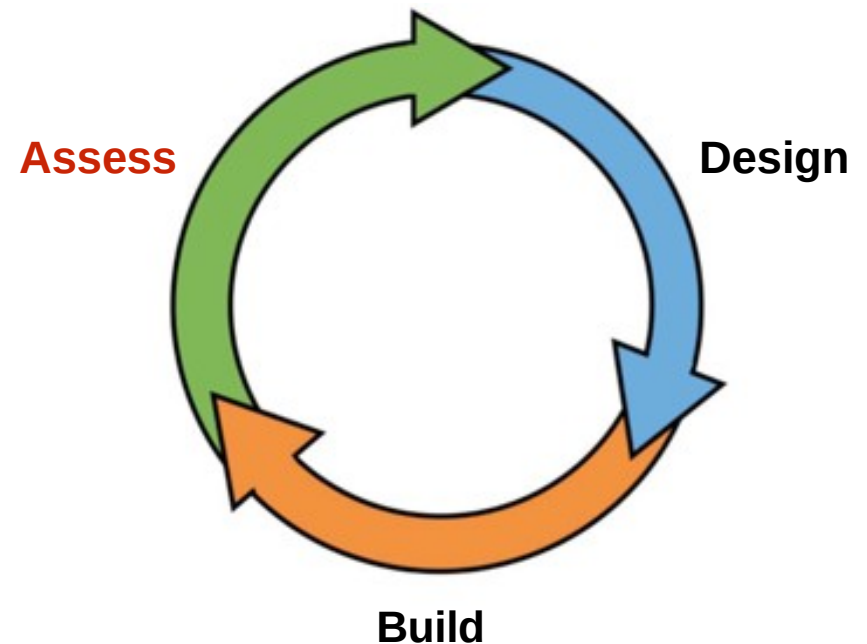
What happens after prototyping?

- **Assessment**

- *assess whether you're on the right track*
- *user testing or through inspection methods or*
- *getting feedback from different stakeholders*

- **Redesign**

- **More Prototyping**



Sketching



What is a Sketch?

- A cheap rapidly constructed representation of an idea.
- Quick
- Timely
- Inexpensive
- Disposable
- Plentiful
- Minimal Detail
- Allow ambiguity

Why Sketch?

- **Reflect**
- **Explore**
- **Communicate**
- **It helps us: Create, Record, Reflect, share, critique, decide**

Why Sketch?

- **Create**

- early ideation
- think openly about ideas and think through them
- force you to visualize how things come together
- brainstorming: generate abundant ideas without worrying about quality
- invent and explore concepts

- **Record**

- ideas you develop and come across,
- archive ideas for later reflection

- **Reflect, share, critique, decide**

- communicate ideas to others
- invite responses, criticisms, and alternatives;
- choose ideas worth pursuing

How to Sketch?

- **Use pencil/paper/whiteboard**
- **Go fast**
- **Dont perfect**
- **Make lots of them**

- **You D O N T have to be 'good at drawing'**

What to Sketch?

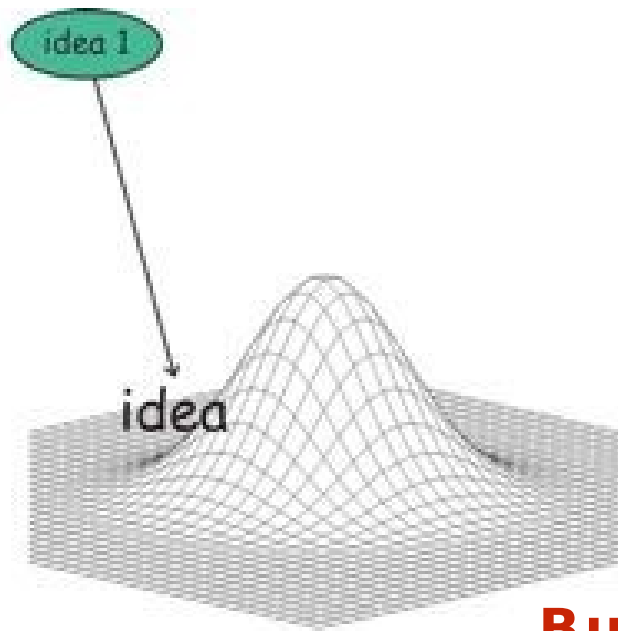
- **The Problem**
- *How would someone experience this problem*
- **The solution**
- *What would it look like for the problem to be solved?*

Getting the Right Design

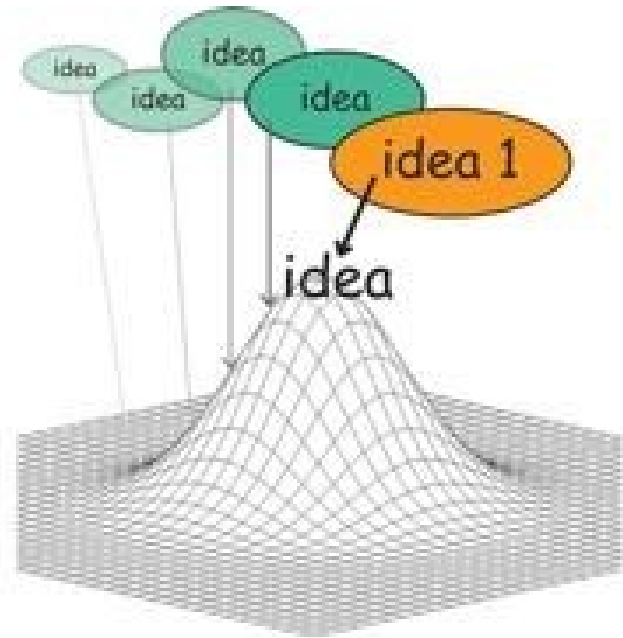
- **Generate many ideas and variations**
- **Reflect on all ideas**
- **Choose the ones that look most promising**
- **Develop them in parallel**
- **Add new ideas as they come up**
- **Then iterate your final choice**

Getting the Right Design

- *Generate an idea*



Iterate and develop it

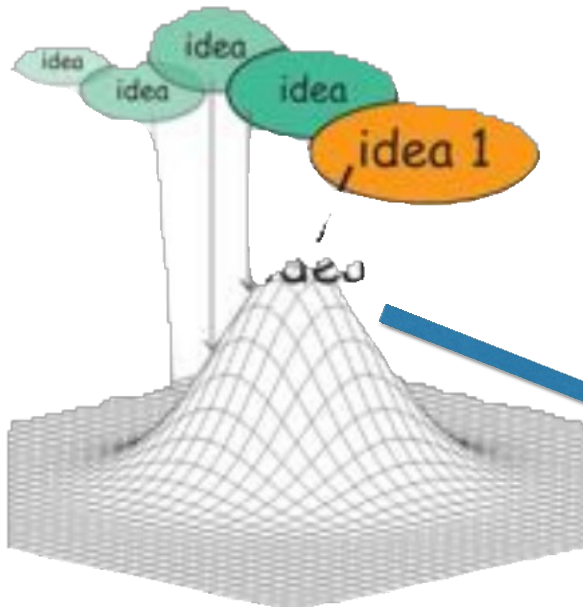


But is it the best idea?

Getting the Right Design

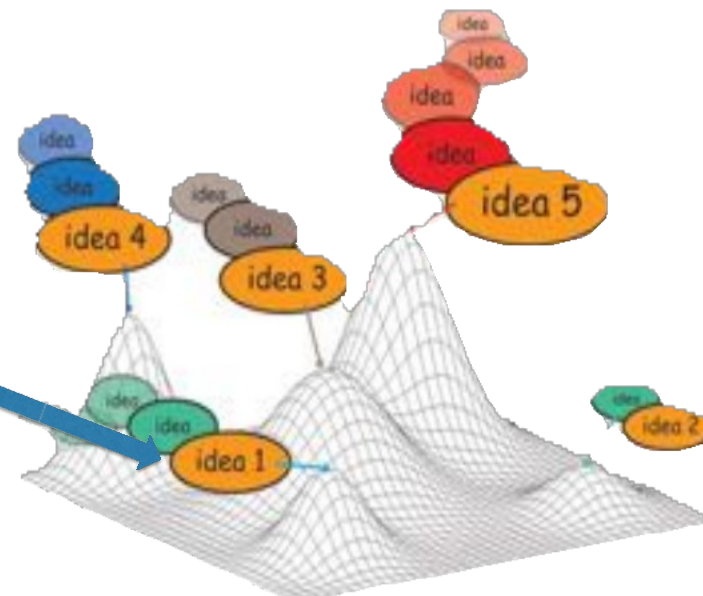
- **The Problem**

- *fixates on first idea*



- **Local hill climbing issue**

- *did you reach local vs. global maxima?*



Opposing Processes in Design

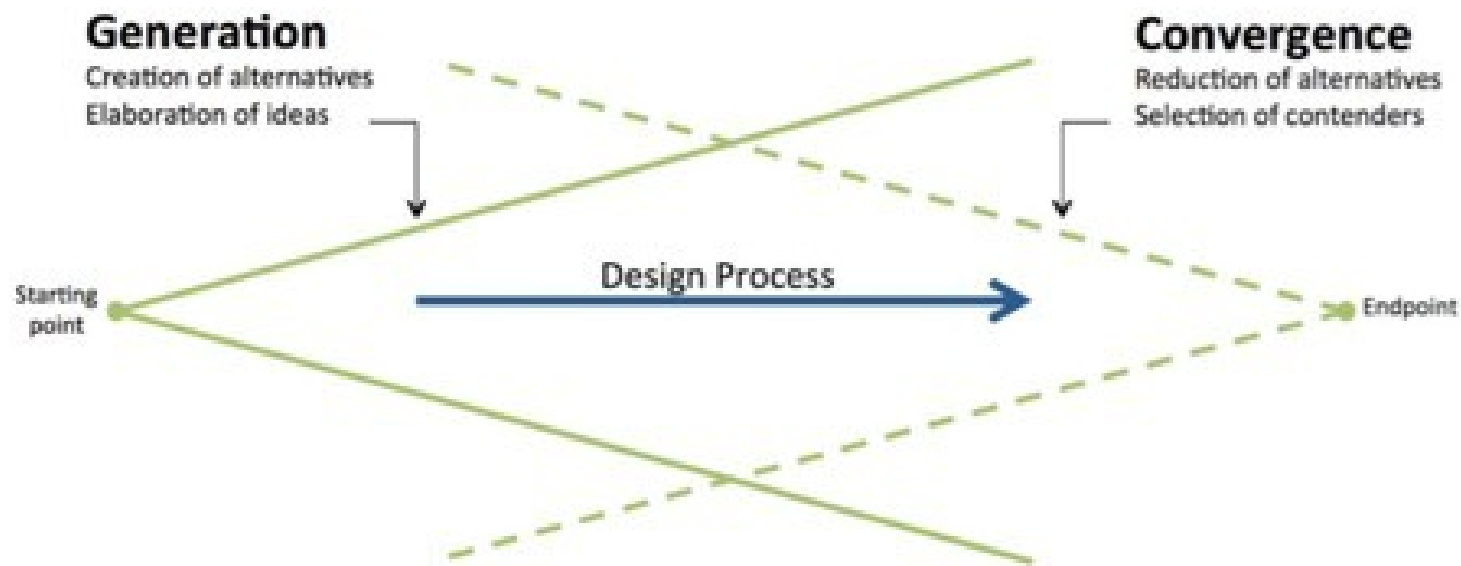


Figure adapted from Buxton, B. Sketching the User Experience

Elaboration and Reduction

- **Elaborate**

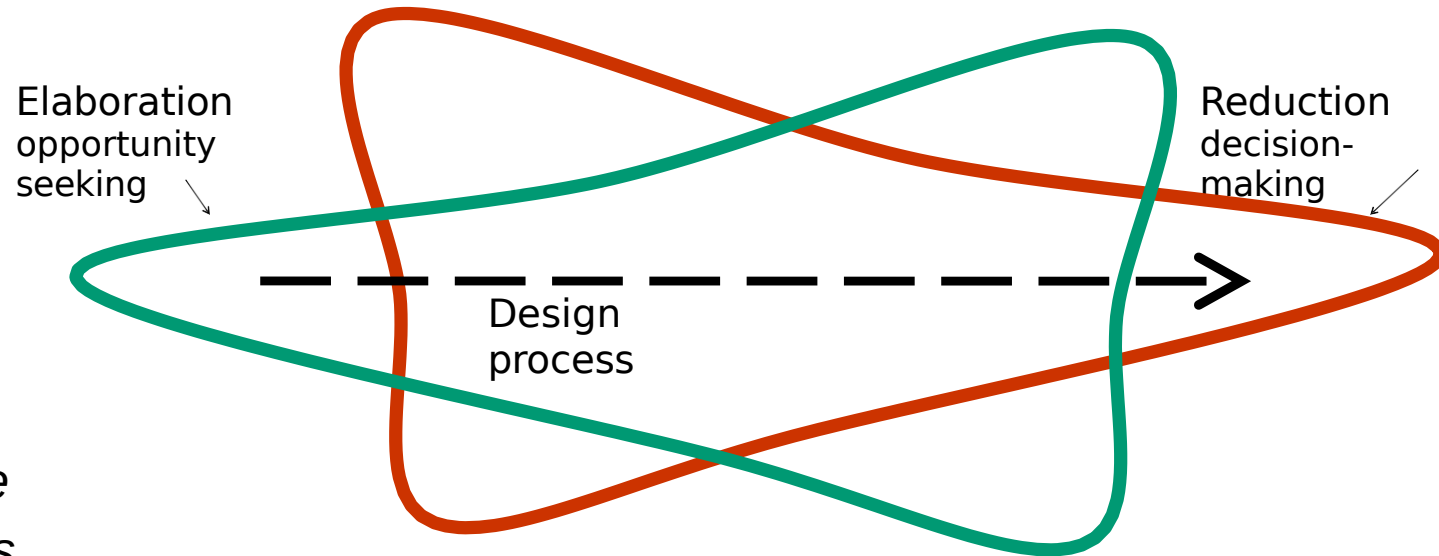
- - *generate solutions.*
These are the opportunities

- **Reduce**

- - *decide on the ones worth pursuing*

- **Repeat**

- - *elaborate and reduce again on those solutions*



Source: Laseau,P. (1980) Graphic Thinking for Architects & Designers. John Wiley and Sons

A Sketch is not a Prototype

Difference is

- **a contrast of purpose (always)**
- **a contrast in form (usually, but not always)**

But

- **it's a continuum**



From Sketches to Prototypes

- **Sketches:**
 - *early ideation stages of design*
- **Prototypes:**
 - *capturing /detailing the actual design*

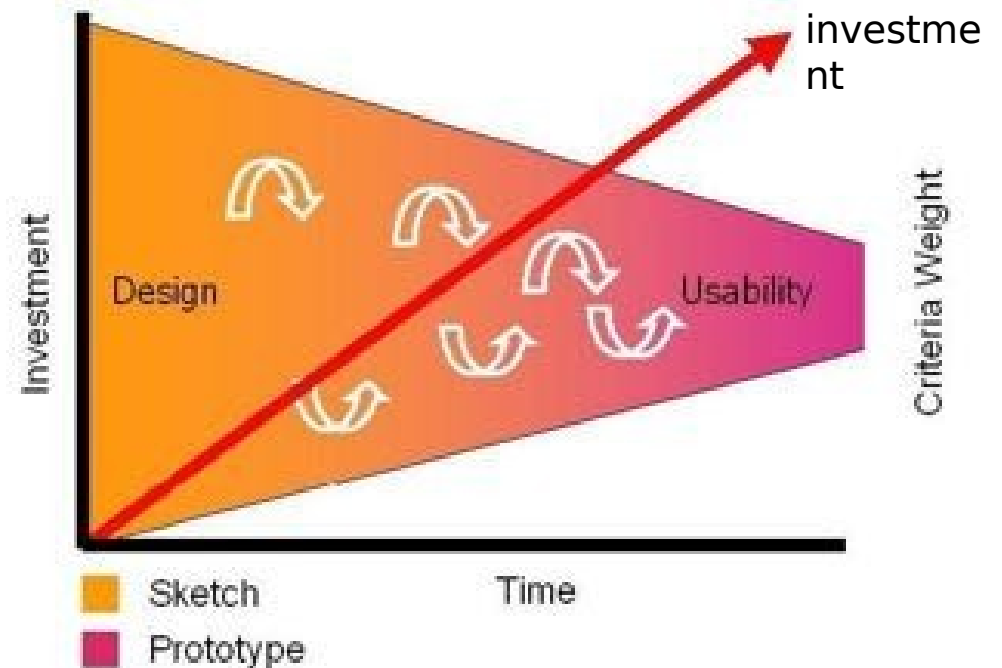
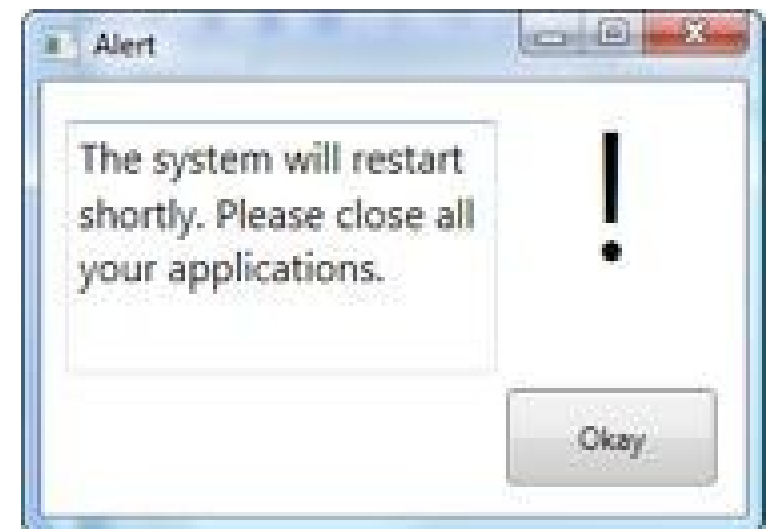
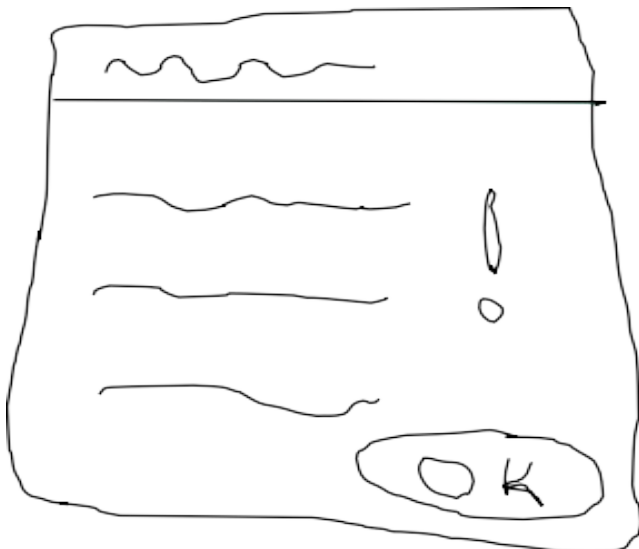


Image from Bill Buxton's Book Sketching User Experiences (2007) Morgan Kaufmann

SKETCH

PROTOTYPE

EVOCATIVE → DIDACTIC excessive instruction
SUGGEST → DESCRIBE
EXPLORE → REFINE
QUESTION → ANSWER
PROPOSE → TEST
PROVOKE → RESOLVE
TENTATIVE → SPECIFIC
NONCOMMITTAL → DEPICTION



Sketching is about design

- **Sketching is not about drawing**
- **It is about design.**
- **Sketching is a tool to help you:**
 - *express*
 - *develop, and*
 - *communicate design ideas*
- **Sketching is part of a process:**
 - *idea generation,*
 - *design elaboration*
 - *design choices,*
 - *engineering*



Summary

